

ECON100B - SECTION 13

Inflation

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ANNOUNCEMENTS/AGENDA

Announcements

- Take Home Midterm “Essay” due Friday, 3/20

Agenda

- Take Home Essay Overview
- Inflation

Question of the Day

What feedback do you have for sections so far?



Any post-midterm questions?

Midterm Essay Overview

- 1-page policy brief, so think more of a Memo compared to a full-length essay
- Builds off the ideas of two Op-Ed articles from the NYT
- A little bit of research may be required, but nothing too heavy
- Goal is to connect the arguments of these Op-Eds into class content
- Feel free to incorporate some of your own opinions and ideas into the essay as well, but focus on primarily answering the questions
- Due next Friday

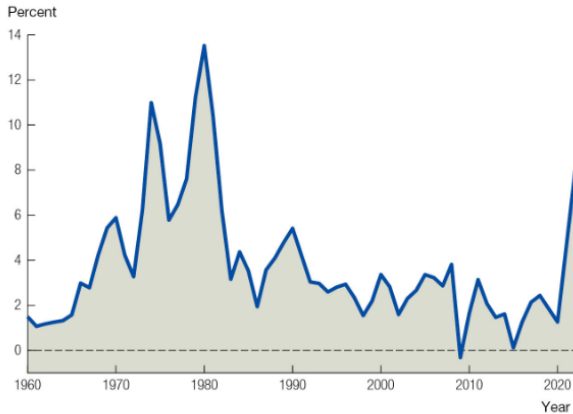
Inflation



Inflation: a Base Overview

- Inflation is defined as the percentage change in an economy's overall price level
- Calculated as: $(P_{t+1} - P_t)/P_t$
 - Periods of extremely high inflation are known as hyperinflation: Think Latin America in the 1980s and 1990s, Russia in the early 1990s, Zimbabwe in the late 2000s, Venezuela in the 2010s/2020s

The Inflation Rate in the United States



Source: Federal Reserve Bank of St. Louis;
fred.stlouisfed.org/series/cpiaucsl.

What even is money?

A recap on money

- “Fiat money” is money that is government issued but not backed up by any commodity
- Money becomes valued largely by social convention and agreement

Measuring the supply of money

- The narrowest measure is physical currency (\$2.3T)
- A broader measure is the “monetary base”, including currency and the accounts that private banks hold with an economy’s central bank, reserves which banks are required to have (\$5.3T)
- An even broader measure is “M1” which includes currency plus demand deposits (checking accounts) (\$19.6T)
- The broadest general measure is “M2” which includes M1 plus savings deposits and individual money market accounts (\$21.3T)

The Quantity Theory of Money

The quantity theory of money states:

$$M_t V_t = P_t Y_t$$

- M represents the amount of “money” in the economy, V represents the velocity of money (think the number of times that a unit of money gets used in a transaction every year), P represents the price level, and Y denotes real GDP
- The right side of our equation represents nominal GDP – the amount of goods and services valued at current prices
- According the quantity theory, the goods and services are bought with money, and the left side of our equation represents the amount of money used in purchased in a given year

The Classical Dichotomy

- States that in the long run, the real and nominal sides of the economy are completely separate
- So, real GDP can be taken as an exogenous variable, determined by what we discussed in the previous growth units
- Also assume the velocity of money is an exogenous constant (relatively constant in empirical settings)
- For now, we will also assume that the money supply is exogenous, but this is a policy lever that can be changed by the central bank of an economy
- Therefore, we can get the following equation for the Price level:

$$P_t^* = \frac{\bar{M}_t \bar{V}}{\bar{Y}_t}.$$

- The key takeaway: in the long run, **a key determinant of the price level is the level of the money supply**

Calculating inflation through growth rates

$$\bar{g}_M + \bar{g}_V = g_P + \bar{g}_Y.$$

- Assume that the velocity is constant
- Also assume that we denote the rate of inflation (or change in price level) as π .

$$\pi^* = \bar{g}_M - \bar{g}_Y.$$

The neutrality of money

- The idea that changes in the money supply have no real effects on the economy and only affect prices
- Empirical evidence suggests that the neutrality of money does not hold in the short run
- Nominal prices do not respond immediately and precisely to changes in the money supply, so it can affect the real side of the economy over short horizons
- Personal argument: the neutrality of money also doesn't consider the psychological impacts of short-term changes in the price level/inflation that can have impacts on the “real” economy

Inflation Practice Questions

- Was inflation high or low during the 1970s? What caused it to be this way, and what policies were used to reduce inflation during this period? What are some of the ripple effects
- Suppose that I am in an economy with a money growth rate of 8%, a constant velocity, and a real GDP growth rate of 3%. What would the inflation rate equal under the quantity theory of money?
- Suppose that I am in an economy that has a money supply of \$500 and a nominal GDP of \$4,000. What would be the velocity of money in this economy?
- If I double my money supply, what does the theory of “The Neutrality of Money” state will happen to my real GDP?

Practice Question Answers

- Causes: Expansionary monetary policy, oil crises, and other causes. Solutions: Volcker Shock, high interest rates to disincentivize debt and spending. Results: More neo-liberal policies, lower inflation for the next period of American history
- $8\% - 3\% = 5\%$. See slides for details
- $4,000/500 = 8$. See slides for details
- The theory states there will be impact in the short-run or long-run. Empirically, we say there is likely a short-run impact (likely negative)

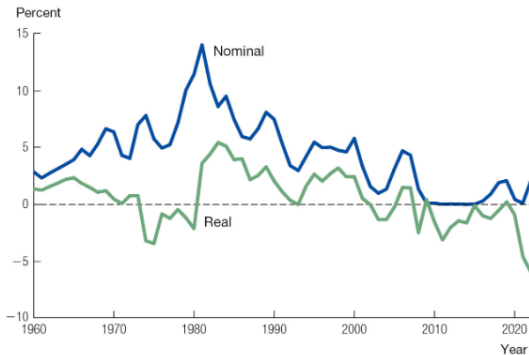
Real and Nominal Interest Rates

- Recall we defined the real interest rate as equal to the MPK, or the return to purchasing a unit of capital, investing it, getting the return to investment, and then selling the capital
- The interest rates that we see in bank accounts and in the news are nominal interest rates
- If we let R denote the real interest rate and i denote the nominal interest rate, we find the Fisher equation:

$$i = R + \pi$$

- We don't directly measure a "real interest rate", but we calculate it by applying the Fisher equation

Real and Nominal Interest Rates in the United States



Source: FRED and author's calculations. The nominal interest rate plotted here is the annualized return on 3-month Treasury bills.

The Costs of Inflation

- Changes the distribution of who is winning and losing
- People with debts they can pay back with cheaper dollars are considered winners, while creditors end up as losers
- Inflation most harmful when unexpected and in environments that are not ready for the change
- Taxes on nominal values can be harmful when real values have not gone up correspondingly
- Inflation also distorts relative prices based on what adjusts to inflation at what speeds
- Other traditional costs of inflation include “shoe-leather costs”, the cost of dealing with things like more banking during periods of inflation, and “Menu costs”, which are costs to firms of changing prices when inflation is high

Short Aside – The Costs of Deflation

- Not in-scope, but why do we have inflation? Why doesn't the central bank just pursue a policy that focuses on zero inflation or even deflation?
- Further reading, Debt-Deflation Theory: Keynes -> Fisher -> Bernanke
- Consider that (for better or worse) that our modern economies are largely built on debt (think student loans, mortgage payments)
- If there was deflation, but interest rates were still positive it would become extremely difficult for people to pay off their debts!
- Leads to defaults on loans, hurting both individuals and banks, then causing bankruptcies for banks, then causing distress for businesses -> perpetuates the cycle
- Fisher's full version of this theory is a little bit more complicated but good reading for those interested in Macro-history and Great Depression History.

Monetary vs. Fiscal causes of inflation

- Consider the government budget constraint:

$$\underbrace{G}_{\text{uses}} = \underbrace{T + \Delta B + \Delta M}_{\text{sources of funds}}$$

- The revenue the government obtains by issuing new money is called seignorage or the inflation tax
- Called the inflation tax because it hurts people holding money to print more money, because now their money is worth less
- In normal times, government will try to finance most if not all of its funding through taxes and debt, but in certain times it may have to resort to printing currency if other avenues become too difficult
- This is the root cause of many of the modern hyperinflation episodes
- This is why central bank independence is considered so important to prevent fiscal considerations from altering monetary policies (!)

Inflation Practice Questions

- Suppose we had a nominal interest rate of 5% and an inflation rate of 2%. What would our real interest rate be under this scenario?
- Who would be the most hurt by inflation under this scenario? An individual who pays a variable rate mortgage on a house or an individual who owns a plot of land?
- Milton Friedman stated, “high inflation is always and everywhere a monetary phenomenon”. Why do you agree or disagree with this statement?

Practice Question Answers

- $5\% - 2\% = 3\%$. See slides for details
- The variable mortgage rate. They have to pay more on the house payments, while the value of the land remains constant
- You may choose to disagree, as we discussed, there are significant effects the fiscal policy can have on monetary policy, hence it can be considered a fiscal problem as well.